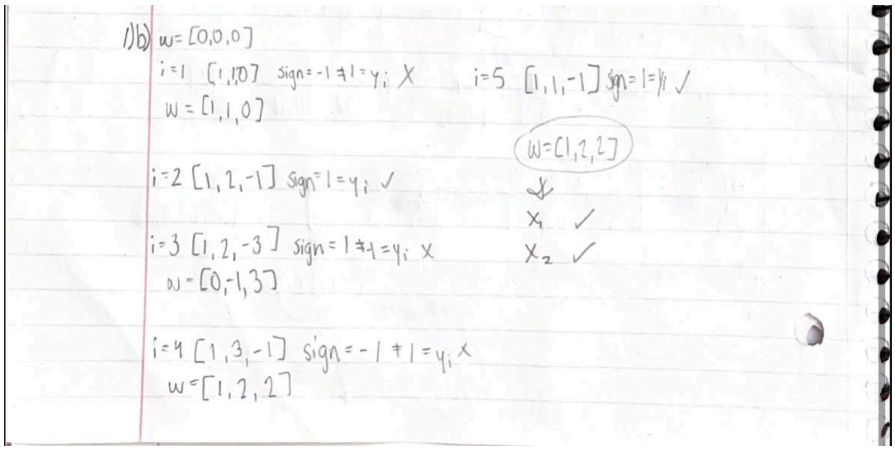
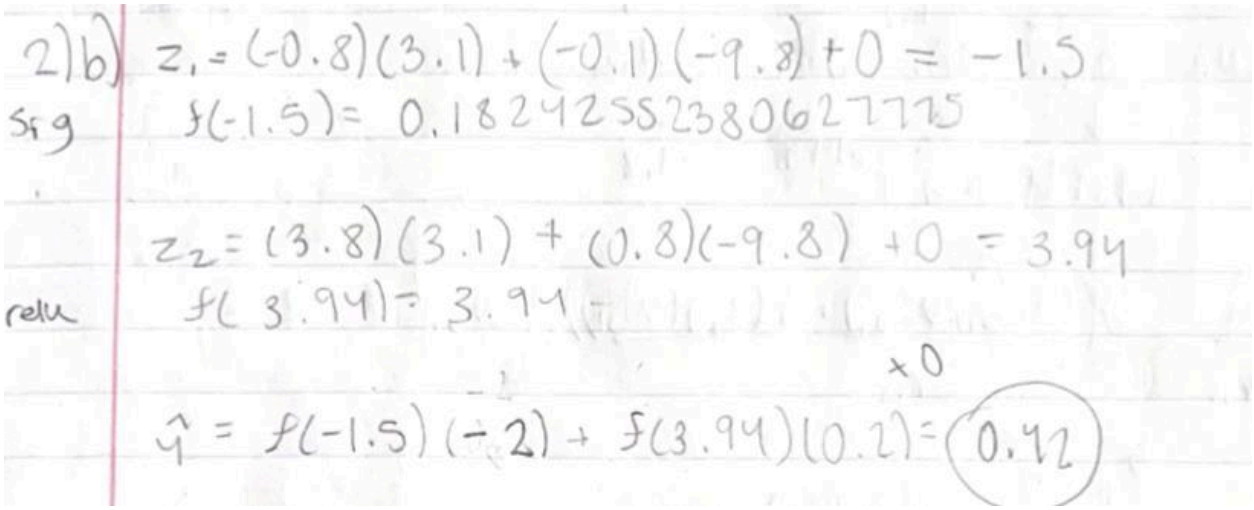
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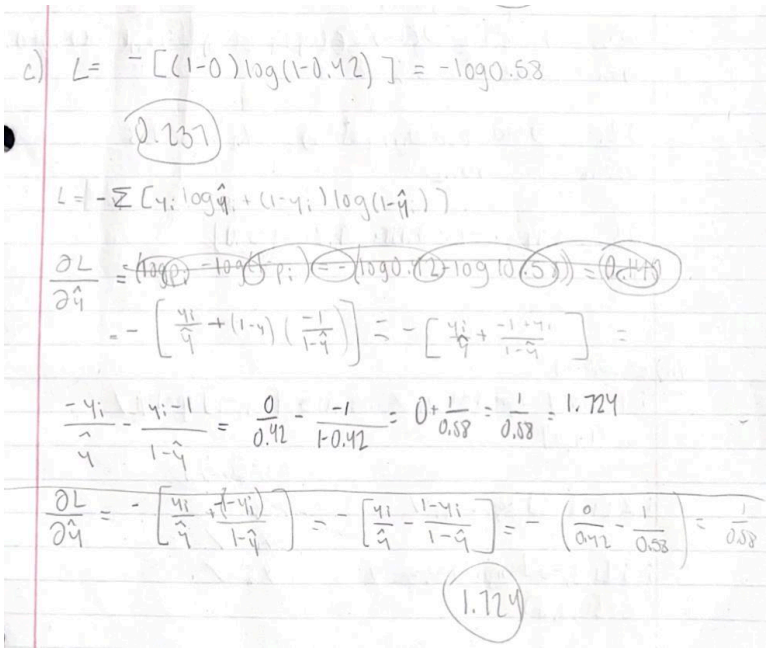
1. Perceptron
    a. We see that the weight vector was updated when i = 1, 2, 4, and 5, but not 3. This means the prediction did not match the true label for those data points, so the weight vector was updated. In this epoch, only when i = 3, or x_3, y_3 in the dataset was classified correctly.
    b. w = [1,2,2] The model's prediction of x_2 will be +1 if we iterate through the dataset again.



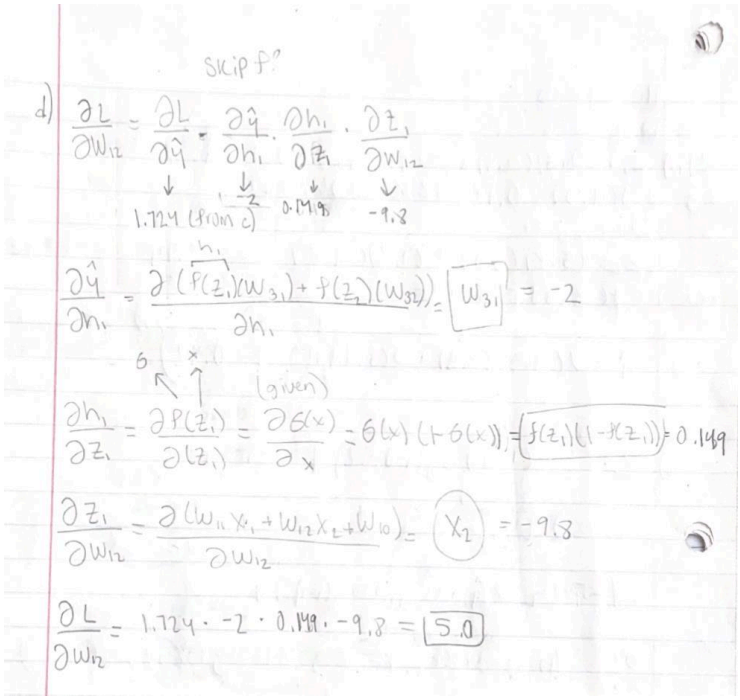
    c. When the data is not linearly separable, the perceptron will never converge. The perceptron will keep infinitely updating the weight vector, or stop at the max number of iterations (if specified), but will never converge to a proper hyperplane that separates the data correctly. The difference is also that perceptron uses the step function, while logistic regression uses sigmoid. On the other hand, logistic regression can find a decision-boundary with non-linearly separable data. This is because it uses minimization of the negative log likelihood function, which is convex, so it will converge to a possible solution.
2. Neural Networks
    a. For the hidden layers, we can use any activation function (that is non-linear) since we don't need a strict output type and need to introduce non-linearity. This can be tanh, ReLU, etc. For the output layer, we can use the sigmoid function because we want binary output since we are doing a binary classification problem. The sigmoid function bounds results between 0 and 1, making it ideal for this case. We can interpret the sigmoid outputs as probabilities and use them to determine how likely the point is to belong to one of the two classes.
    b. 0.42



c. L = 0.237 ; 1.724

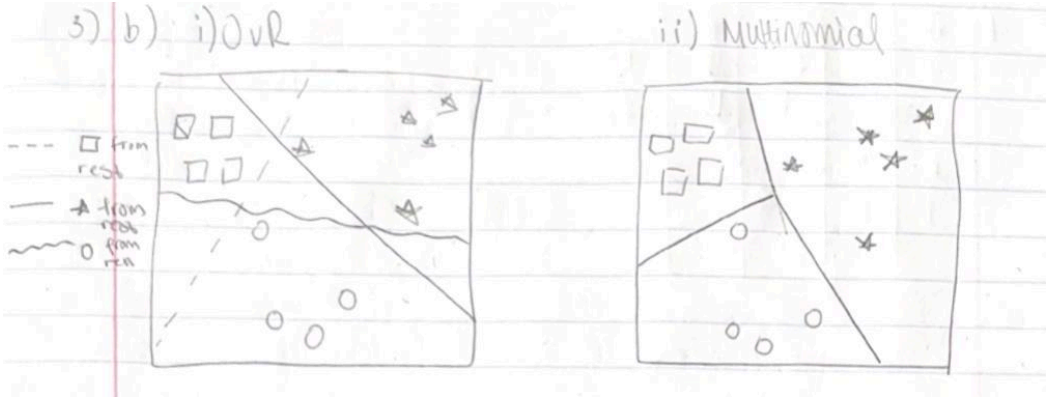


d. 5.0 ; Since we have a positive partial derivative and want to minimize the loss function, we should decrease the bias to decrease loss.



e. There are 9 parameters. There are 6 weight parameters (2 from x_1 in input layer, 2 from x_2 in input layer, 1 from w_1 in hidden layer, 1 from w_2 in hidden layer) and 3 bias terms (W_{10}, W_{20}, and W_{30}). This results in 6 weight terms + 3 bias terms = 9 parameters.

3. Multi-Class Classification (4 classes, 25 features)
    a. If we used OVR strategy with 4 classes and 25 features, we would have 4 logistic regressions, one for each class. In each logistic regression, we would have one parameter for bias and then one for each of the 25 features. 4*(1+25) = 104 parameters.
    b.



4. Decision Boundary
    a. Plot A matches 2. Compared to ReLU, tanh results in more smooth, curved decision boundaries (seen below from lecture slides). When combining 10 of these layers, we get an overall smooth boundary like we see in plot A.
    b. Plot B matches 1. ReLU results in a sharper decision boundary as can be seen by the graph below from class. ReLU would look like putting together 10

piecewise linear functions that are less smooth than in tanh, resulting in a sharper decision boundary overall, like we see in plot B.

